

ECO362 Midterm Solutions

Narek Alexanian, Anshu Chen, Jiwon Lee, Jialiang Lin, Jihong Song

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Q1

(a) Notations:

- Let $c_{2020}(T)$ be the T -year par yield in 2020.
- Let $P_{b,2020}(T)$ be the price of a T -year zero coupon bond with face value \$1 in 2020.

We use the formula for 1-year par yield:

$$c_{2020}(1) = \frac{1 - P_{b,2020}(1)}{P_{b,2020}(1)} \quad (1)$$

After reorganizing (1), we get:

$$P_{b,2020}(1) = \frac{1}{1 + c_{2020}(1)} = \frac{1}{1 + 0.12\%} = \frac{1}{1 + 0.0012} \approx 0.9988 \quad (2)$$

Thus the price of a 1-year zero coupon bond with \$100 face value is: $100P_{b,2020}(1) \approx \boxed{99.88}$

Grading:

- 5 points for the fully correct answer.
 - For full credit, the answer must **either** explicitly derive the relationship between $P_{b,2020}(1)$ and $c_{2020}(1)$ as above, **or** state clearly that 1-year zero coupon bond yield equals 1-year par yield $c_{2020}(1) = Y_{b,2020}(1)$.
- 2 points for any numerically correct answer that **either** treats 1-year par yield as 1-year zero coupon bond yield without any justification, **or** plugs 0.12% into the price-yield relationship without stating where the number 0.12% comes from.
 - *Example:* an answer that only states $P_{b,2020}(1) = 1/[1 + Y_{b,2020}(1)] = 1/[1 + 0.0012] = 0.9988$ will receive 2 points.
 - Such an answer only qualifies for partial credit, because there is no evidence that the student understands the difference between par yields and zero-coupon bond yields.
- We deduct 0.5 point for (1) any calculation or transcription error, (2) using continuous compounding, (3) using the wrong yield from the table, or (4) neglecting the \$100 face value.

(b) We use the formula for 2-year par yield:

$$c_{2020}(2) = \frac{1 - P_{b,2020}(2)}{P_{b,2020}(1) + P_{b,2020}(2)} \quad (3)$$

After reorganizing 3, we get:

$$P_{b,2020}(2) = \frac{1 - c_{2020}(2) \times P_{b,2020}(1)}{1 + c_{2020}(2)} = \frac{1 - 0.0012 \times 0.9988}{1 + 0.0012} \approx 0.9976 \quad (4)$$

Thus the price of a 2-year zero coupon bond with **\$100** face value is: $100P_{b,2020}(1) \approx \boxed{99.76}$

Grading:

- 5 points for the fully correct answer.
 - For full credit, the answer must derive 2-year zero coupon bond price based on the 2-year par yield equation (3) or its equivalent.
- 2 points for any numerically correct answer that either explicitly treats the 2-year par yield as 2-year zero coupon bond yield (it is wrong), or implicitly does so by plugging 0.12% directly into the price-yield relationship for 2-year zero coupon bond without stating where the number comes from.
 - *Example:* an answer that states $P_{b,2020}(2) = 1/[1 + Y_{b,2020}(2)]^2 = 1/[1 + 0.0012]^2 = 0.9976$ will receive 2 points.
 - Such an answer only qualifies for partial credit, because there is no evidence that the student understands the difference between par yields and zero-coupon bond yields.
- We deduct 0.5 point for (1) any calculation or transcription error, (2) using continuous compounding, (3) using the wrong yield from the table, or (4) neglecting the \$100 face value.

(c) Based on the table, the 3-year par bond you buy in 2019 has coupon rate of $c_{2019}(3) = 1.64\%$, and the purchasing price is \$100.

In 2020, this bond becomes a *1.64% coupon bond with 2-year remaining maturity*. Let $c = 1.64\%$ denote the coupon rate. We can compute the selling price based on the present value formula for coupon bond:

$$P_c(2) = 100 \cdot c \cdot P_{b,2020}(1) + 100 \cdot (1 + c) \cdot P_{b,2020}(2) \quad (5)$$

$$= 100 \cdot 0.0164 \cdot 0.9988 + 100 \cdot 1.0164 \cdot 0.9976 \approx 103.03 \quad (6)$$

Finally, we compute the (annually compounded) return of this investment.

$$r = \frac{\text{Selling Price} + \text{Coupon}}{\text{Purchasing Price}} \quad (7)$$

$$= \frac{P_c(2) + 100 \cdot c}{100} \quad (8)$$

$$= \frac{103.03 + 100 \cdot 0.0164}{100} = \boxed{4.67\%} \quad (9)$$

Grading: Grades are awarded incrementally by step.

- (i) +1.5 point for writing down the correct equation for 1-year investment return.
 - Either equation (7) or its continuously compounded version is accepted. Must include coupon payment in the equation.
 - (ii) +1 point for recognizing the purchasing price is equal to the face value.
 - (iii) +1.5 point for recognizing that the bond becomes a 2-year *coupon bond* in 2020, and writing down the present value formula of a 2-year coupon bond (equation 5) for computing the selling price.
 - (iv) +0.5 point for plugging in correct numbers into the present value formula for computing the selling price.
 - (v) +0.5 point for the correct numerical answer, awarded only if all previous steps are correct.
- (d) Let $P_{a,2021}(3)$ denote the price of a 3-year annuity with face value \$1 in 2021. The present value formula for annuities provides:

$$P_{a,2021}(3) = P_{b,2021}(1) + P_{b,2021}(2) + P_{b,2021}(3) \quad (10)$$

We use the same method as in part (a) to compute the 1-year zero coupon bond price in 2021:

$$P_{b,2021}(1) = \frac{1}{c_{2021}(1)} = \frac{1}{1 + 0.0042} \approx 0.9958 \quad (11)$$

We use the same method as in part (b) to compute the 2-year zero coupon bond price in 2021:

$$P_{b,2021}(2) = \frac{1 - c_{2021}(2) \cdot P_{b,2021}(1)}{1 + c_{2021}(2)} = \frac{1 - 0.0076 \cdot 0.9958}{1 + 0.0076} \approx 0.9849 \quad (12)$$

To compute the 3-year zero coupon bond price, we start from the 3-year par yield equation, and reorganize it a bit:

$$c_{2021}(3) = \frac{1 - P_{b,2021}(3)}{P_{b,2021}(1) + P_{b,2021}(2) + P_{b,2021}(3)} \quad (13)$$

$$= \frac{1 - P_{a,2021}(3) + P_{b,2021}(1) + P_{b,2021}(2)}{P_{a,2021}(3)} \quad (14)$$

Finally, we reorganize (14) to get an expression for $P_{a,2021}(3)$, and plug in numbers from equations (11) and (12) to get the final answer.

$$P_{a,2021}(3) = \frac{1 + P_{b,2021}(1) + P_{b,2021}(2)}{1 + c_{2021}(3)} = \frac{1 + 0.9958 + 0.9849}{1 + 0.0099} \approx \boxed{2.95} \quad (15)$$

Grading:

- 5 points for the fully correct answer.

- For full credit, the answer must derive zero coupon bond prices / annuity prices *from par yield*.
- 2 points for any numerically correct answer that either explicitly treats the par yields as zero coupon bond yields (it is wrong), or implicitly does so by using the par yields from the table in a price-yield relationship without any justification.
 - *Example:* an answer that states $P_a(3) = \frac{1}{1+0.0042} + \frac{1}{(1+0.0076)^2} + \frac{1}{(1+0.0099)^3} = 2.95$ will receive 2 points.
 - Such an answer only qualifies for partial credit, because there is no evidence that the student understands the difference between par yields and zero-coupon bond yields.
- We separately deduct 0.5 point for any calculation or transcription error, or for using continuous compounding, or for using a wrong face value.
- We separately deduct 0.5 point for any answer that uses the wrong yield from the table (e.g. doing all calculation based on the 3-year par yield).

Special notes:

- Any answer that writes down the present formula for annuity with incorrect or no further development will not receive partial credit.
- Any answer that writes down a wrong present value formula (e.g. the one for coupon bonds instead of annuities) will not receive partial credit.

(e) The present value of a 3-year annuity that pays \$350000 per year is:

$$350000 \times P_{a,2021}(3) = 350000 \times 2.95 = 1032500 \quad (16)$$

Therefore, you should take the 3-year annuity because it has higher **present value** than the one-time payment of \$1 million.

Grading:

- 5 points for correct reasoning based on the annuity price computed in part (d).
 - To receive full credit, the answer must explicitly calculate the **present value** for the annuity option based on the answer in part (d), and compare this **present value** against the amount of one-time payment.
 - An answer that gets the wrong conclusion only due to a mistake in part (d) will still receives full credit.
- 0 point for any answer with incorrect or no reasoning.
 - *Example 1:* an answer that states the value of annuity as $3 \times 350000 = 1050000$ will receive 0 point: such a reasoning is incorrect because it doesn't consider discounting. As a general rule, you should never add money from different years together without proper compounding / discounting.
 - *Example 2:* an answer that only states "annuity has higher present value than one-time payment" without any calculation will also receive 0 point.
- If a student provides no numerical answer for part (d), then an answer for part (e) only qualifies for 5 point if it satisfies the following two criteria:

- (i) Explicitly states the present value logic for making the comparison.
- (ii) Explicitly states that the answer for part (e) will depend on the 3-year annuity price asked by part (d).

Q2

- (a) The price of the riskless asset is $1/1.01 = \$0.9901$. By using this riskless asset and Disney, we can construct a replicating portfolio of Verizon.

$$1w_{rf} + 145w_{Disney} = 46$$

$$1w_{rf} + 88w_{Disney} = 34$$

Solve for $w_{rf} = 15.47368, w_{Disney} = 0.21053$.

By assuming no arbitrage,

$$0.9901w_{rf} + 101w_{Disney} = 36.58$$

The current price of Verizon is \$36.58.

Grading:

2 points for the correct equations for the replicating portfolio.

1 point for numerically correct quantities for the replicating portfolio.

1 point for writing the correct equation for the security's price.

1 point for the numerically correct price.

- (b) We can also construct the replicating portfolio of this security by holding Disney and the riskless asset.

$$1w_{rf} + 145w_{Disney} = 1$$

$$1w_{rf} + 88w_{Disney} = 0$$

Solve for $w_{rf} = -1.54386, w_{Disney} = 0.01754$.

Then the price of the security that pays \$1 in expansion and \$0 in recession is

$$0.9901w_{rf} + 101w_{Disney} = 0.243$$

Grading:

2 points for the correct equations for the replicating portfolio.

(1 point if the replicating portfolio is constructed with Disney and Verizon without using the riskless asset)

1 point for numerically correct quantities for the replicating portfolio.

1 point for writing the correct equation for the security's price.

1 point for the numerically correct price.

(c) Likewise,

$$1w_{rf} + 145w_{Disney} = 0$$

$$1w_{rf} + 88w_{Disney} = 1$$

Solve for $w_{rf} = 2.54386$, $w_{Disney} = -0.01754$.

Then the price of the security that pays \$0 in expansion and \$1 in recession is

$$0.9901w_{rf} + 101w_{Disney} = 0.747$$

Grading:

2 points for the correct equations for the replicating portfolio.

(1 point if the replicating portfolio is constructed with Disney and Verizon without using the riskless asset)

1 point for numerically correct quantities for the replicating portfolio.

1 point for writing the correct equation for the security's price.

1 point for the numerically correct price.

(d) By no arbitrage, the price of a portfolio holding (b) and (c) which mimics the payoffs of Verizon in each state should be equal to the price of Verizon. Thus,

$$P_{Verizon} = X_{Verizon,Expansion}P_1 + X_{Verizon,Recession}P_2$$

$$\Rightarrow 36.58 = 46 \times 0.243 + 34 \times 0.747$$

Grading:

5 points for the correct answer.

4 points for numerical errors with the correct equation.

2 points for one key error in the equation.

1 point for two key errors in the equation.

Q3

Asset	Conservative	Moderate	Aggressive
US stocks	17.5%	28%	35%
UK stocks	15%	24%	30%
Japanese stocks	7.5%	12%	15%
German stocks	10%	16%	20 %
T-bills	50%	20%	0

The solution of this exercise depends on **the separation theorem**; the risky portfolio of all agents has the same proportions of the underlying assets. What investor differ in is their exposure to this risky portfolio.

For simplicity assume that everyone has 100 dollars to invest.

1. Aggressive investor holds only risky assets. Her portfolio is the tangency portfolio. The holdings of risky assets by the other agents will mimick the proportions, she chooses and will be scaled down versions of her choice.
2. The conservative investor invests 50 dollars in risky assets (100 -50); she needs to put 35% of that in the US stocks (mimicking the aggressive investor); hence, $50 \cdot 0.35 = 17.5$ dollars; and $50 \cdot 0.15 = 7.5$ dollars in the Japanese stocks, meaning 7.5 %. That leaves her $50 - (17.5 + 7.5) = 15$ dollars to put in the UK stock market.
3. The German stocks are $10/50 = 20\%$ of the conservative investor. Hence, the aggressive investor needs to put the same proportion, $0.2 \cdot 100 = 20$ dollars; 20%. Going back to the aggressive investors, since her total wealth in the risky assets is double that of the conservative investor, she needs to put 20 dollars in the risky asset. That leaves her with 30 dollars to allocate to the UK stocks.
4. Lastly, for the moderate investor we see that she puts 24% in the UK stocks, which has to be 30% of her wealth. Hence, she puts 80 dollars in the risky assets. That leaves her with $100 - 30$ dollars for the riskless. US stocks will be $0.35 \cdot 80 = 28$ and $0.15 \cdot 80 = 12$ for Japanese stocks. This leaves $80 - (24 + 12 + 28) = 16$ dollars or 20% of 80.

Grading:

3 points were deducted for each mistaken gap;

carry-over mistakes were penalized once;

3 additional points(additional mistake) were deducted if column entries did not sum to 100(except if that was due to rounding)

Note: Rounding corrections propagated across the box leading to answers that were far from the answers without rounding; these deviations were not penalized

Q4

- (a) What is the expected excess return (i.e., $\mathbb{E}[R_{i,t} - R_b]$) on Amazon? What is the expected excess return on BP?

The expected excess return of stock i is given by

$$\mathbb{E}[R_{i,t} - R_b] = \beta_i \mathbb{E}[R_{m,t} - R_b] = \beta_i (\mathbb{E}[R_{m,t}] - R_b). \quad (17)$$

Hence, for Amazon, it is

$$0.87(0.94\% - 0.08\%) = 0.75\%, \quad (18)$$

and for BP, it is

$$1.53(0.94\% - 0.08\%) = 1.32\%. \quad (19)$$

Grading:

5 points for correct formula and numerical results.

4 points for correct formula with computational mistakes only.

2 points for correct formula.

- (b) What are the mean and the standard deviation of returns on a portfolio that invests 50% in each of Amazon and BP?

Approach 1: Derive the statistical model for the portfolio return from models of individual stocks:

$$R_{pt} - R_b = (.5\beta_{Amazon} + .5\beta_{BP})(R_{mt} - R_b) + .5\epsilon_{Amazon,t} + .5\epsilon_{BP,t} \quad (20)$$

Therefore, the mean return of the portfolio is

$$\mathbb{E}[R_{pt}] = R_b + (.5\beta_{Amazon} + .5\beta_{BP})(\mathbb{E}[R_{mt}] - R_b) \quad (21)$$

$$= 0.08\% + 0.5(0.87 + 1.53)(0.94\% - 0.08\%) \quad (22)$$

$$= 1.11\%, \quad (23)$$

where the second equality uses the fact that $\mathbb{E}[\epsilon_{i,t}] = 0$.

And the standard deviation of the portfolio return is

$$\sigma(R_{pt}) = \sqrt{\text{Var}((.5\beta_{Amazon} + .5\beta_{BP})(R_{mt} - R_b) + .5\epsilon_{Amazon,t} + .5\epsilon_{BP,t})} \quad (24)$$

$$= \sqrt{(.5\beta_{Amazon} + .5\beta_{BP})^2 \text{Var}(R_{mt}) + .5^2 \text{Var}(\epsilon_{Amazon,t}) + .5^2 \text{Var}(\epsilon_{BP,t})} \quad (25)$$

$$= \sqrt{(.5 \times .87 + .5 \times 1.53)^2 (.053)^2 + .5^2 (.075)^2 + .5^2 (.045)^2} \quad (26)$$

$$= 0.0772 \text{ or } 7.72\%, \quad (27)$$

where the second equality uses the fact that $\text{Cov}(R_{mt}, \epsilon_{it}) = 0$.

Approach 2: Write the portfolio return in terms of stock returns:

$$R_{pt} = .5R_{Amazon,t} + .5R_{BP,t}. \quad (28)$$

Then the mean return of the portfolio is

$$\mathbb{E}[R_{pt}] = .5\mathbb{E}[R_{Amazon,t}] + .5\mathbb{E}[R_{BP,t}] \quad (29)$$

$$= .5(R_b + \mathbb{E}[R_{Amazon,t} - R_b]) + .5(R_b + \mathbb{E}[R_{BP,t} - R_b]) \quad (30)$$

$$= .5[0.08\% + 0.7482\%] + .5[0.08\% + 1.3158\%] \quad (31)$$

$$= 1.11\% \quad (32)$$

where the third equality uses results from part (a). And the standard deviation of portfolio return is

$$\sigma(R_{pt}) = \sqrt{\text{Var}(.5R_{Amazon,t} + .5R_{BP,t})} \quad (33)$$

$$= \sqrt{.5^2 \text{Var}(R_{Amazon,t}) + .5^2 \text{Var}(R_{BP,t}) + 2 \times .5 \times .5 \times \text{Cov}(R_{Amazon,t}, R_{BP,t})} \quad (34)$$

$$= 0.0772 \text{ or } 7.72\%, \quad (35)$$

where

$$Var(R_{Amazon,t}) = \beta_{Amazon}^2 Var(R_{mt}) + Var(\epsilon_{Amazon,t}) \quad (36)$$

$$= .87^2 \times .053^2 + .075^2 \quad (37)$$

$$Var(R_{BP,t}) = \beta_{BP}^2 Var(R_{mt}) + Var(\epsilon_{BP,t}) \quad (38)$$

$$= 1.53^2 \times .053^2 + .045^2 \quad (39)$$

$$Cov(R_{Amazon,t}, R_{BP,t}) = Cov(\beta_{Amazon}R_{mt} + \epsilon_{Amazon,t}, \beta_{BP}R_{mt} + \epsilon_{BP,t}) \quad (40)$$

$$= \beta_{Amazon}\beta_{BP}Var(R_{mt}) \quad (41)$$

$$= .87 \times 1.53 \times .053^2. \quad (42)$$

One could also write the above equations in matrix forms:

$$\mathbb{E}[R_{pt}] = (w_{Amazon} \ w_{BP}) \begin{pmatrix} \mathbb{E}[R_{Amazon,t}] \\ \mathbb{E}[R_{BP,t}] \end{pmatrix} \quad (43)$$

$$\sigma(R_{pt}) = \sqrt{(w_{Amazon} \ w_{BP}) \begin{pmatrix} Var(R_{Amazon,t}) & Cov(R_{Amazon,t}, R_{BP,t}) \\ Cov(R_{Amazon,t}, R_{BP,t}) & Var(R_{BP,t}) \end{pmatrix} \begin{pmatrix} w_{Amazon} \\ w_{BP} \end{pmatrix}} \quad (44)$$

Grading:

5 points for correct formulas and numerical results.

4 points for correct formulas for both mean and standard deviation with only computational mistakes.

3 points for correct formula for either mean or standard deviation.

- (c) What are the mean and the standard deviation of returns on a portfolio that invests 30% in Amazon and 70% in BP?

Derive the statistical model for the portfolio return from models of individual stocks:

$$R_{pt} - R_b = (.3\beta_{Amazon} + .7\beta_{BP})(R_{mt} - R_b) + .3\epsilon_{Amazon,t} + .7\epsilon_{BP,t} \quad (45)$$

Therefore, the mean return of the portfolio is

$$\mathbb{E}[R_{pt}] = R_b + (.3\beta_{Amazon} + .7\beta_{BP})(\mathbb{E}[R_{mt}] - R_b) \quad (46)$$

$$= 0.08\% + (.3 \times .87 + .7 \times 1.53)(0.94\% - 0.08\%) \quad (47)$$

$$= 1.22\%, \quad (48)$$

where the second equality uses the fact that $\mathbb{E}[\epsilon_{i,t}] = 0$.

And the standard deviation of the portfolio return is

$$\sigma(R_{pt}) = \sqrt{Var((.3\beta_{Amazon} + .7\beta_{BP})(R_{mt} - R_b) + .3\epsilon_{Amazon,t} + .7\epsilon_{BP,t})} \quad (49)$$

$$= \sqrt{(.3\beta_{Amazon} + .7\beta_{BP})^2 Var(R_{mt}) + .3^2 Var(\epsilon_{Amazon,t}) + .7^2 Var(\epsilon_{BP,t})} \quad (50)$$

$$= \sqrt{(.3 \times .87 + .7 \times 1.53)^2 (.053)^2 + .3^2 (.075)^2 + .7^2 (.045)^2} \quad (51)$$

$$= 0.0805 \text{ or } 8.05\%, \quad (52)$$

where the second equality uses the fact that $Cov(R_{mt}, \epsilon_{it}) = 0$.

Grading:

5 points for correct formulas and numerical results.

4 points for correct formulas for both mean and standard deviation with only computational mistakes.

3 points for correct formula for either mean or standard deviation.

- (d) Which of the two portfolios in parts (b) and (c) cannot be the tangency portfolio? State your reasoning clearly for full credit.

Calculate the Sharpe ratios for both portfolios.

50/50 portfolio:

$$SR = \frac{0.0111 - 0.0008}{0.0772} = 0.1334$$

30/70 portfolio:

$$SR = \frac{0.0122 - 0.0008}{0.0805} = 0.1416$$

Since the 30/70 portfolio has a higher Sharpe ratio, the 50/50 cannot be the tangency portfolio.

Grading:

5 points for correct answer with explanation about the Sharpe ratio or some intuitive explanation, i.e., pointing out that the 30/70 portfolio has a somewhat higher expected return but a much higher standard deviation.

3 points for correct answer with a vague explanation that is not incorrect.

2.0 for correct answer with no explanation.

- (e) Is the minimum variance portfolio equal to a portfolio that invests 100% in Amazon and 0% in BP? State your reasoning clearly for full credit.

Answer: No, it is not. Several possible ways to get this answer.

- It is possible to see, by plugging the parameters into the following equation and comparing with the other portfolios in (c) and (d), that we can achieve a lower variance.

$$Var(R_p) = [w\beta_i + (1-w)\beta_j]^2\sigma(R_{mt})^2 + w^2\sigma(\epsilon_{it})^2 + (1-w)^2\sigma(\epsilon_{jt})^2$$

- Other counterexample of a portfolio with lower variance.
- Calculate the actual weights for the minimum variance portfolio, described below. The weight on Amazon comes out to **0.3946**, and BP is **0.6054**.
- Intuitive explanation: since Amazon and BP are not perfectly correlated, it is possible to diversify idiosyncratic risk by holding a mix of the two.

Common Mistakes: Comparing Amazon and BP's idiosyncratic standard deviations and concluding that the answer is "no" because Amazon's is higher than BP's. This receives 0 points.

Common Mistakes: Similarly, no points for calculating Amazon and BP's return standard deviations and concluding that the answer is "yes" because Amazon's standard deviation is lower than BP's.

Grading:

5 points for correct answer with reasoning: counterexample or intuitive explanation about diversification.

3 points for correct answer with a vague explanation that is not incorrect.

2.0 for correct answer with no explanation.

To calculate the minimum variance portfolio weights, we first write out the portfolio variance and then set up the first-order condition to find the optimal weights. Let R_p equal the portfolio return.

$$\begin{aligned} R_p - R_b &= [w\beta_i + (1-w)\beta_j](R_{mt} - R_b) + w\epsilon_{it} + (1-w)\epsilon_{jt} \\ \text{Var}(R_p) &= [w\beta_i + (1-w)\beta_j]^2\sigma(R_{mt})^2 + w^2\sigma(\epsilon_{it})^2 + (1-w)^2\sigma(\epsilon_{jt})^2 \\ &= \sigma(R_{mt})^2(w^2\beta_i^2 + (1-w)^2\beta_j^2 + 2w(1-w)\beta_i\beta_j) + w^2\sigma(\epsilon_{it})^2 + (1-w)^2\sigma(\epsilon_{jt})^2 \end{aligned}$$

Take FOC:

$$\begin{aligned} \sigma(R_{mt})^2(2w\beta_i^2 - 2(1-w)\beta_j^2 + 2\beta_i\beta_j(1-2w)) + 2w\sigma(\epsilon_{it}) - 2(1-w)\sigma(\epsilon_{jt}) &= 0 \\ 2\sigma(R_{mt})^2(w\beta_i^2 - (1-w)\beta_j^2 + \beta_i\beta_j(1-2w)) + 2w\sigma(\epsilon_{it}) + 2w\sigma(\epsilon_{jt}) - 2\sigma(\epsilon_{jt}) &= 0 \\ \sigma(R_{mt})^2(w\beta_i^2 - (1-w)\beta_j^2 + \beta_i\beta_j(1-2w)) + w\sigma(\epsilon_{it}) + w\sigma(\epsilon_{jt}) - \sigma(\epsilon_{jt}) &= 0 \\ w(\sigma(R_{mt})^2(\beta_i - \beta_j)^2 + \sigma(\epsilon_{it}) + \sigma(\epsilon_{jt})) = \sigma(\epsilon_{jt}) - \sigma(R_{mt})^2(\beta_i\beta_j - \beta_j^2) & \\ w = \frac{\sigma(\epsilon_{jt}) - \sigma(R_{mt})^2(\beta_i\beta_j - \beta_j^2)}{\sigma(R_{mt})^2(\beta_i - \beta_j)^2 + \sigma(\epsilon_{it}) + \sigma(\epsilon_{jt})} & \end{aligned}$$

At this stage, plug in the values given in the problem, and you can choose for w to be the weight on Amazon or BP. The weight on Amazon comes out to **0.3946**, and BP is **0.6054**.